

- 14** Since the Earth has a larger mass than the Moon, the point at which gravitational field strength is zero (i.e. net gravitational force acting on a unit mass) should be closer to the Moon than the Earth, and not at the midpoint P. This also means that at P, the gravitational field points towards the Earth, as the gravitational force acting by the Earth is larger than that by the Moon at that point.

Since gravitational potential is a scalar, it is only zero at infinity, and is always negative within the gravitational field of the two bodies.

Answer: C

$$GM/m$$